DSGE Model Estimation via VAR IRF Matching: The Short and Long of It

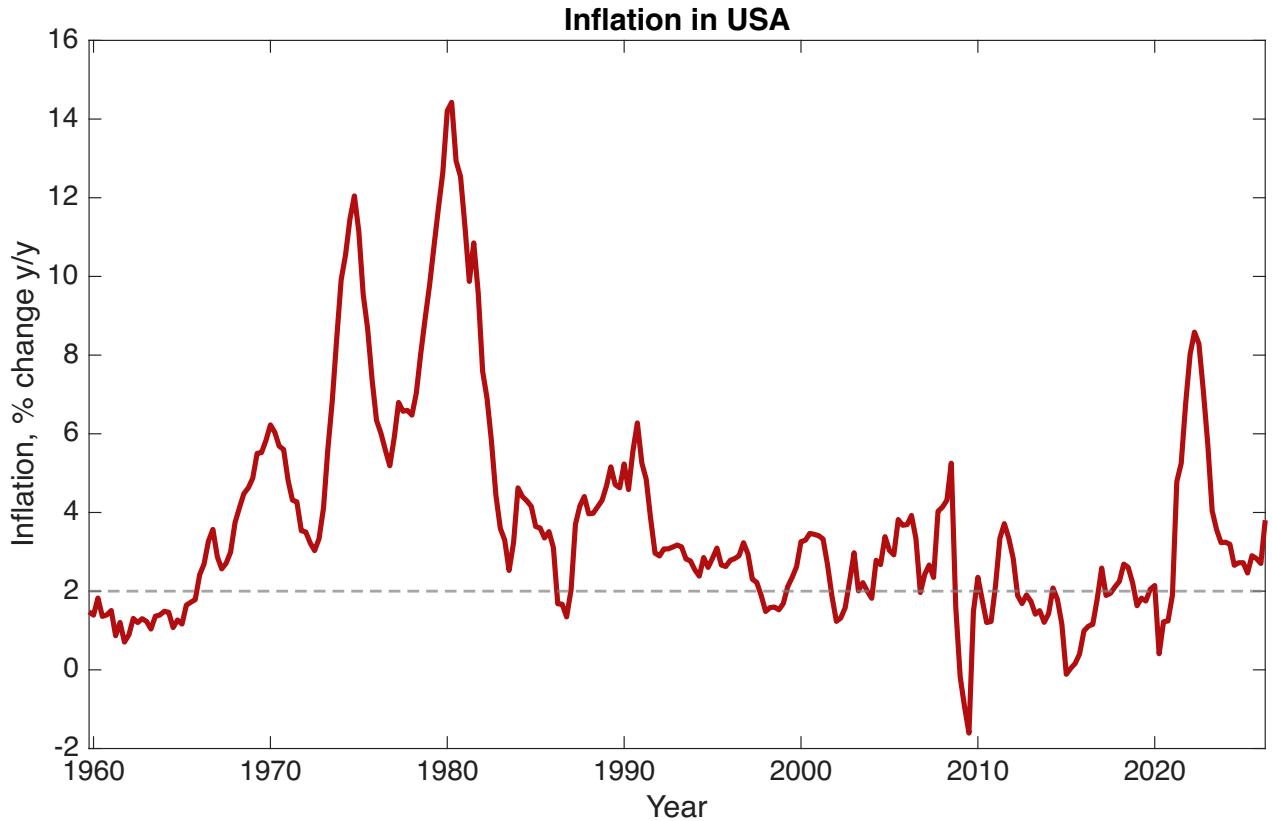
Lorenzo Del Col

Supervisor: Professor Efrem Castelnuovo

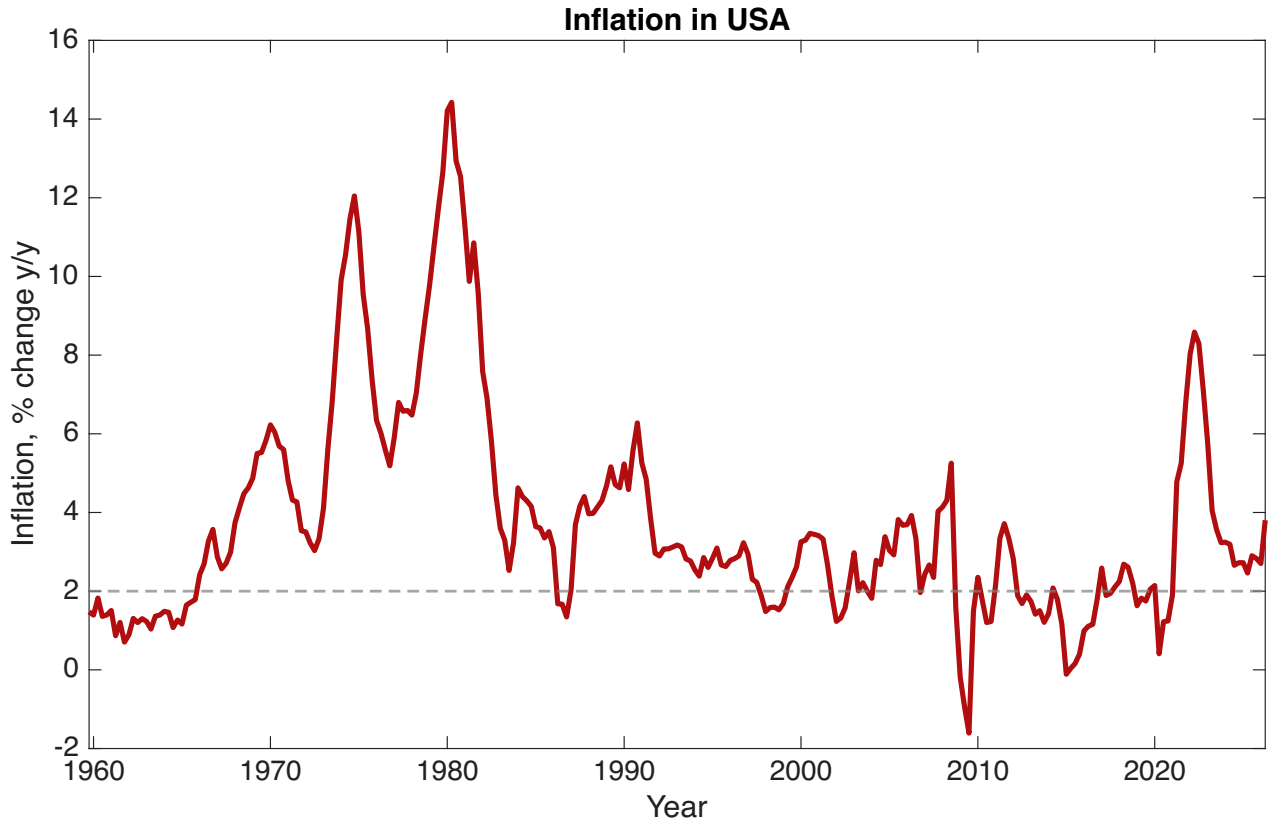
University of Padova
MSc Applied Economics

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Stylised Fact: Inflation Persistence

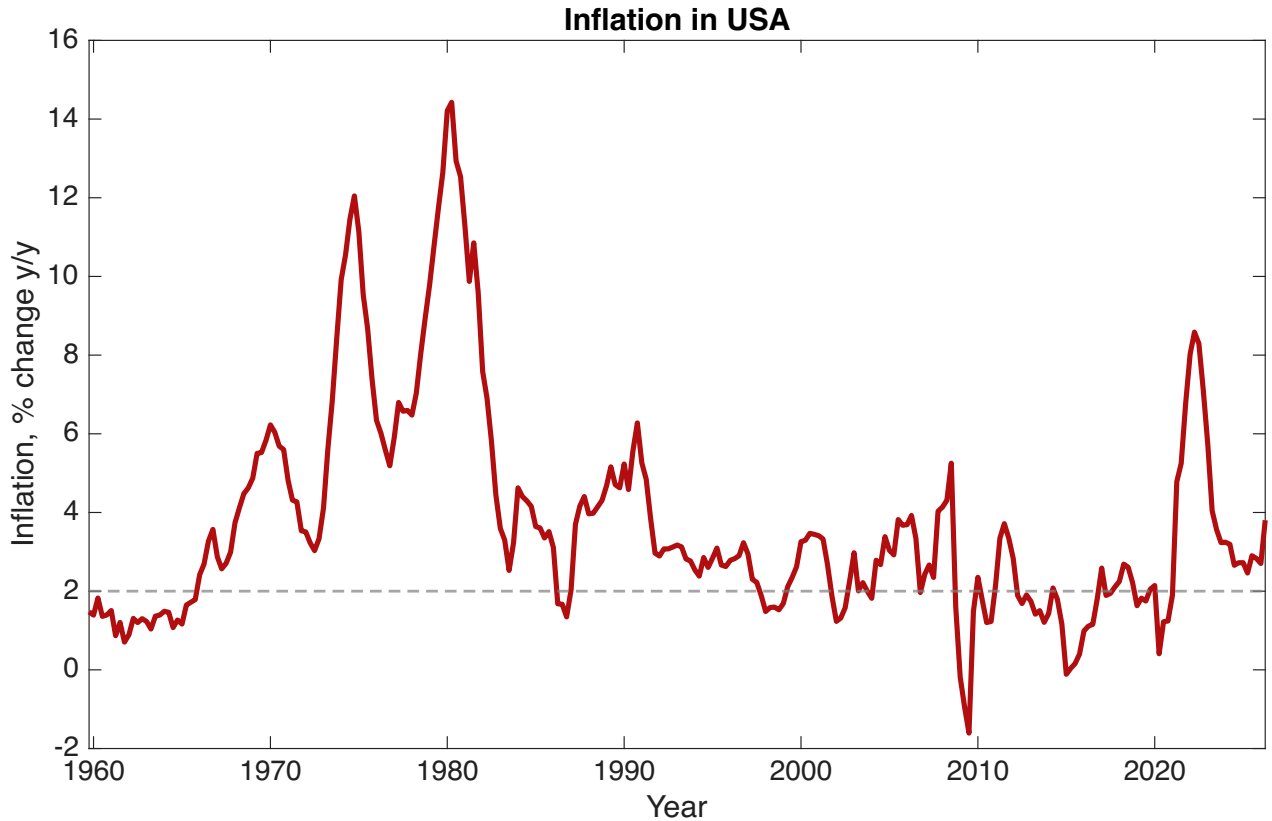


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Stylised Fact: Inflation Persistence



- NK DSGE models need to replicate inflation persistence
- But they often do it assuming **unrealistically high levels of price stickiness**, inconsistent with micro data evidence

CEE and ACEL's DSGE Motivation

- CEE JPE (2005) and ACEL RED (2011) seminal works introduced the concept of **Firm-Specific Capital** to model inflation persistence, mechanism supported by IRF matching with VAR(4)

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- **Impulse Response Function Matching Estimation (IRFME)** is a common way to estimate DSGE models through VARs

$$\hat{\zeta} = \arg\min_{\zeta} (\hat{\Psi} - \Psi(\zeta))' V^{-1} (\hat{\Psi} - \Psi(\zeta)) \tag{1}$$

Motivation

- *"If the* ***VAR is misspecified****, then, the* ***specification errors will be compounded at each horizon****"*

Ramey (2016) p. 83

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- *"If the* ***VAR is misspecified****, then, the* ***specification errors will be compounded at each horizon****"*

  Ramey (2016) p. 83

- **Truncation Bias is a type of misspecification** that arises when the VAR lag order is shorter than its DGP.

- Misspecification can deliver distorted results misinforming the central banker

This Paper

Research Question:

- Can VAR **truncation bias** have significant implications for DSGE estimation?

What's been done:

- Revisit ACEL contribution, estimating **homogeneous** vs **firm-specific capital** versions of their DSGE with VARs with different dynamic structures

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What's been done:

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Findings:

- Increasing the VAR lag structure casts **doubt** on the importance of **specificity of capital** to explain inflation inertia
- Lag choice might have implications for optimal policy design, modelling and structural inference

Related Literature

Price Stickiness:

- Altig et al., (2011); Bils and Klenow, (2004); Golosov and Lucas, (2007); Klenow and Kryvtsov, (2008); Nakamura and Steinsson, (2008)

Long-Lag VARs:

- De Graeve and Westermark, (2025); Baumeister, (2025); Antolín-Díaz and Surico, (2025); Antolín-Díaz et al., (2026); Plagborg-Møller and Wolf, (2021); Ludwig, (2026)

Truncation Bias and the DSGE-VAR mapping:

- Ravenna, (2007); De Graeve and Westermark, (2025); Chari et al., (2008); Barnichon et al., (2026); Fernández-Villaverde et al., (2007); Faust and Leeper, (1997); Lippi and Reichlin, (1993); Forni et al., (2016)

Impulse Response Matching and DSGE Estimation:

- Rotemberg and Woodford, (1997); Christiano et al., (2005); Altig et al., (2011); Iacoviello, (2005); Boivin and Giannoni, (2006); Hall et al., (2012); Giacomini, (2013); Theodoridis, (2011); Guerrón-Quintana, (2010)

ACEL's DSGE Mechanism

ACEL builds two models on Christiano et al. JPE, (2005):

- Homogeneous capital
- **Firm-specific capital**

- The key assumption is that, in any given period, a **aggregate capital stock is pre-determined and firm-specific**
- When firms re-optimise, **prices are adjusted by a small amount** because each firm's short-run marginal cost is increasing in its own output
- **Inflation is inertial** despite firms re-optimising prices more frequently than in the homogeneous capital model

ACEL's DSGE Mechanism (Cont'd)

- When log-linearised, the only difference in the characterising equilibrium equations lies in the Phillips Curve

$$\Delta\hat{\pi}_t = \mathbb{E}[\beta\Delta\hat{\pi}_{t+1} + \gamma\hat{s}_t|\Omega_t] \tag{2}$$

$$\gamma = \frac{(1-\xi_p)(1-\beta\xi_p)}{\xi_p}\chi \tag{3}$$

- **Models are observationally equivalent in terms of prices and quantities:** macro data identify γ, the difference between the two models lies in the implied ξ_p
- $\hat{s}_t$ is the economy-wide marginal cost, γ the Phillips Curve slope, ξ_p the Calvo price stickiness parameter
- χ reduces the sensitivity of inflation to marginal costs in the firm-specific capital model

ACEL's DSGE Mechanism (Cont'd)

Here χ is a non-linear function representing the strategic complementarity in price setting

- In the **Homogeneous** model, $\chi \equiv 1$
- In the **Firm-specific** model, $\chi < 1$ and captures two forces that make a firm move its price *less* when it can reset:
  - It depends on capacity utilisation cost σ_a
    The costlier it is to vary capital utilisation, the steeper the firm's own marginal cost curve
  - The more elastic the firm's demand, the larger the fall in output (and hence in marginal cost) after a price increase

- χ governs **how much** a firm moves its price
- ξ_p **how often**, with implied duration $1/(1-\xi_p)$

Given a fixed value of γ, a smaller χ implies a smaller ξ_p

ACEL's MP shock propagation mechanism

ACEL (2005) Results

The estimated γ is 0.040, which implies:

- **Homogeneous** capital: $\xi_p = 0.82$, firms change prices every 5.6 quarters
- **Firm-specific** capital: firms change prices every 1.5 quarters

Only the firm-specific capital model reconciles the micro-macro pricing conflict

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- **Are these results robust to an enriched lag structure of the VAR?**

Illustration of the problem: Truncation Bias

$$Y_t = B_1 Y_{t-1} + \cdots + B_p Y_{t-p} + u_t, \qquad \mathbb{E}(u_t u_t') = \Sigma \tag{4}$$

$$B(L) = B_1 L + \cdots + B_p L^p \tag{5}$$

- The VAR with $p < p^*$ suffers from truncation bias.[1]
- Lag truncation biases the reduced-form coefficients $B(L)$ and the reduced-form variance-covariance matrix Σ, which are the two main ingredients for IRF estimation.

$$\text{IRF} = f\big(B(L), \Sigma\big) \tag{6}$$

[1] Where p^* denotes the DGP true lag length.

Long-Lag VARs

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Lower Bias

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Lower Bias

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Lower Variance

- Misspecification reduction lowers the variance of the residuals
- This variance reduction counteracts the imprecision caused by increased parameterisation

Long-Lag VARs Benefits

The variance-reducing effect of lower misspecification often dominates the penalty of increased parameterisation.

Compared to short-lag VARs, Long-Lag VARs have:

- Reduced truncation bias
- More precise inference
- Lower MSE
- Better coverage

Residuals' Variance Reduction

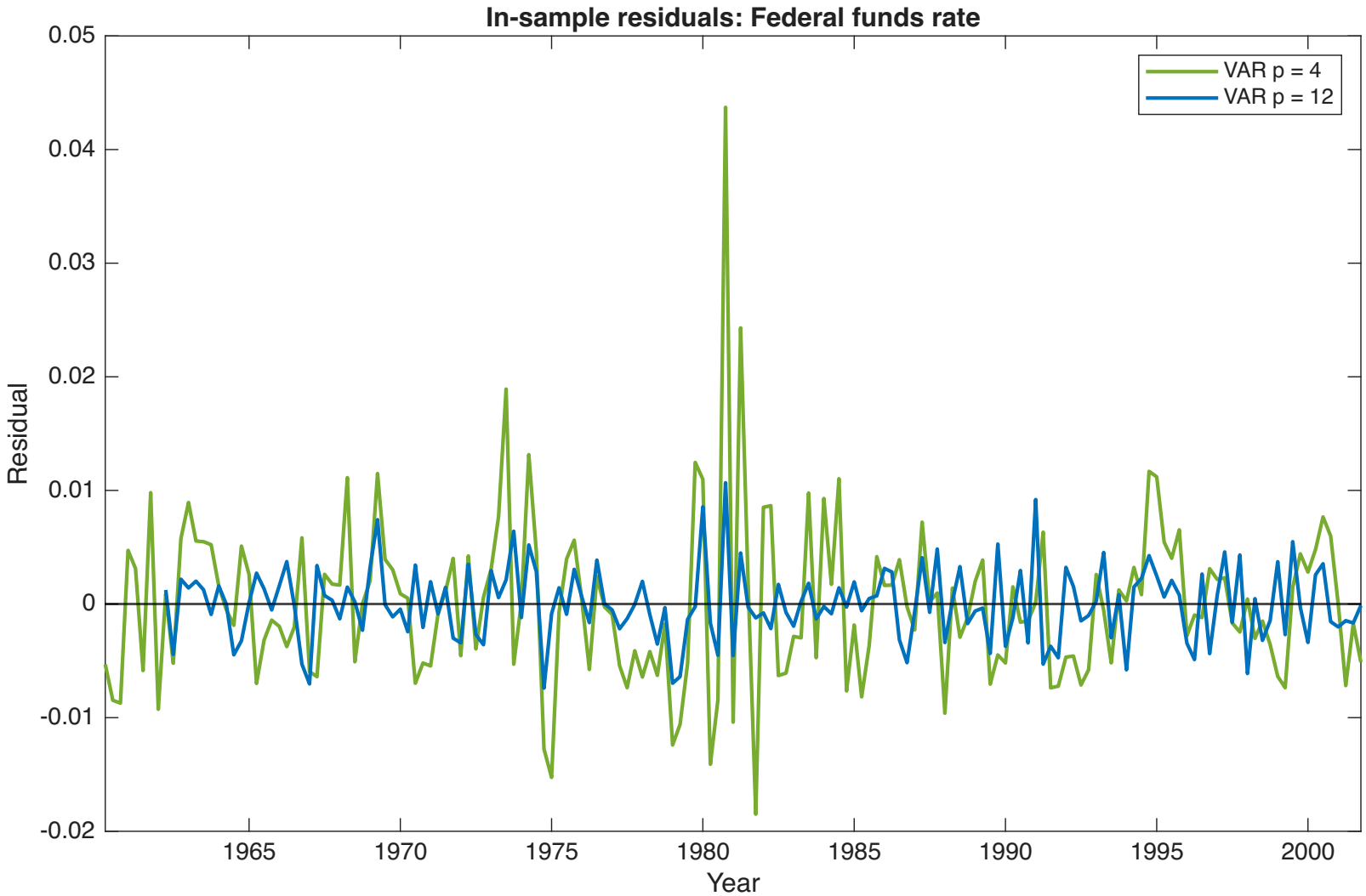


Figure 1: FFR Residuals Comparison Across Lags 4 vs 12

Replication Exercise

Framework:

- **Altig et al. (2011)**'s DSGE[2]

VAR:

- Short and long-run restrictions, medium-scale VAR with 10 variables
- Monetary Policy, Neutral and Embodied Technology shocks
- Sample period 1959:Q2-2001:Q4

Exercise:

- I re-estimate the VAR increasing the lag order
- Re-estimate the DSGE by IRF matching
- **Analyse implications of possible variations in the structural parameters and IRFs across different lag choices**

[2]Where the replication package provided by the authors refers to the 2005 NBER working paper

VAR IRFs; Monetary Policy Shock

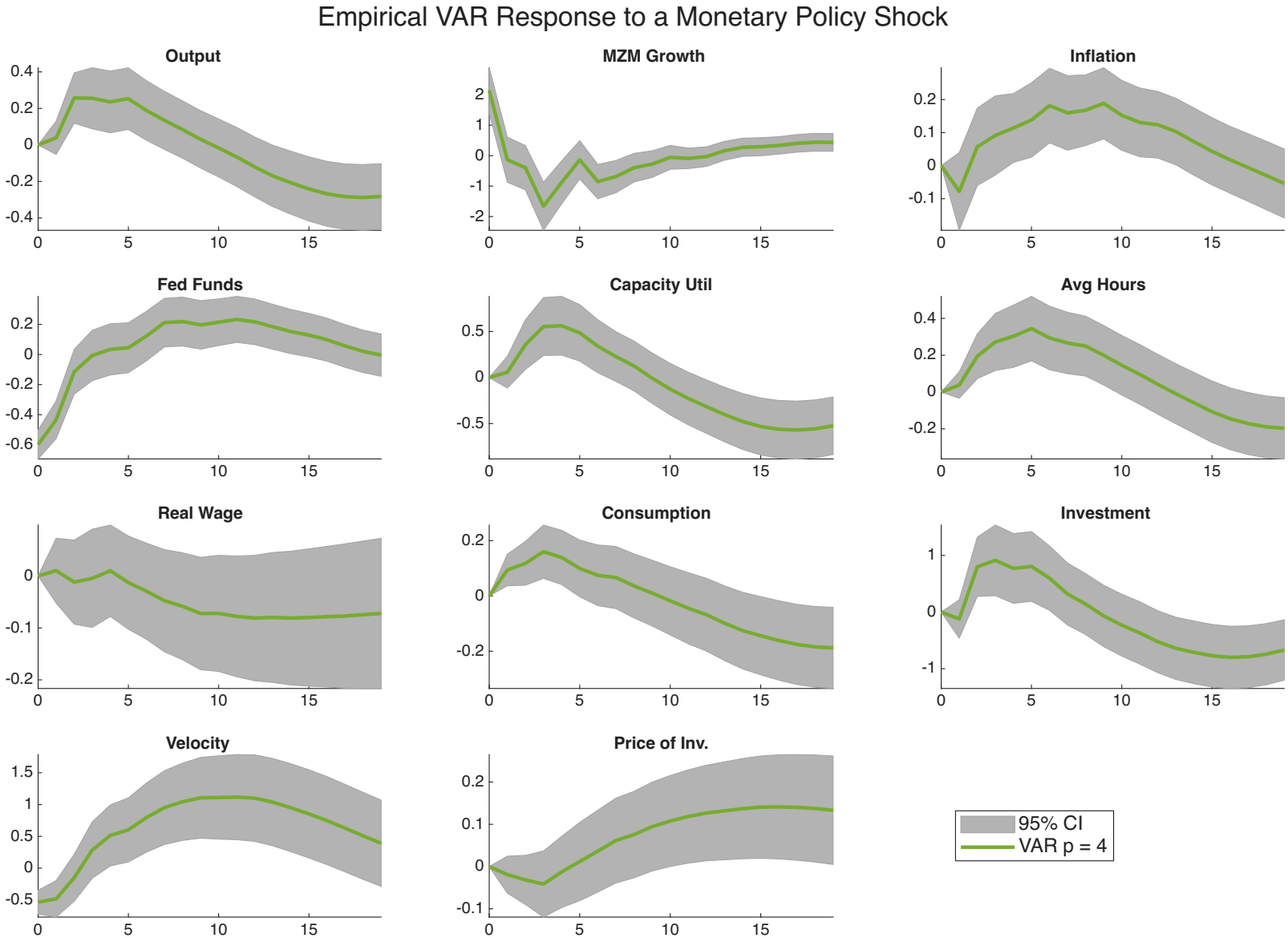


Figure 2: VAR IRFs Comparison: Monetary Policy Shock: 4 lags

VAR IRFs; Monetary Policy Shock

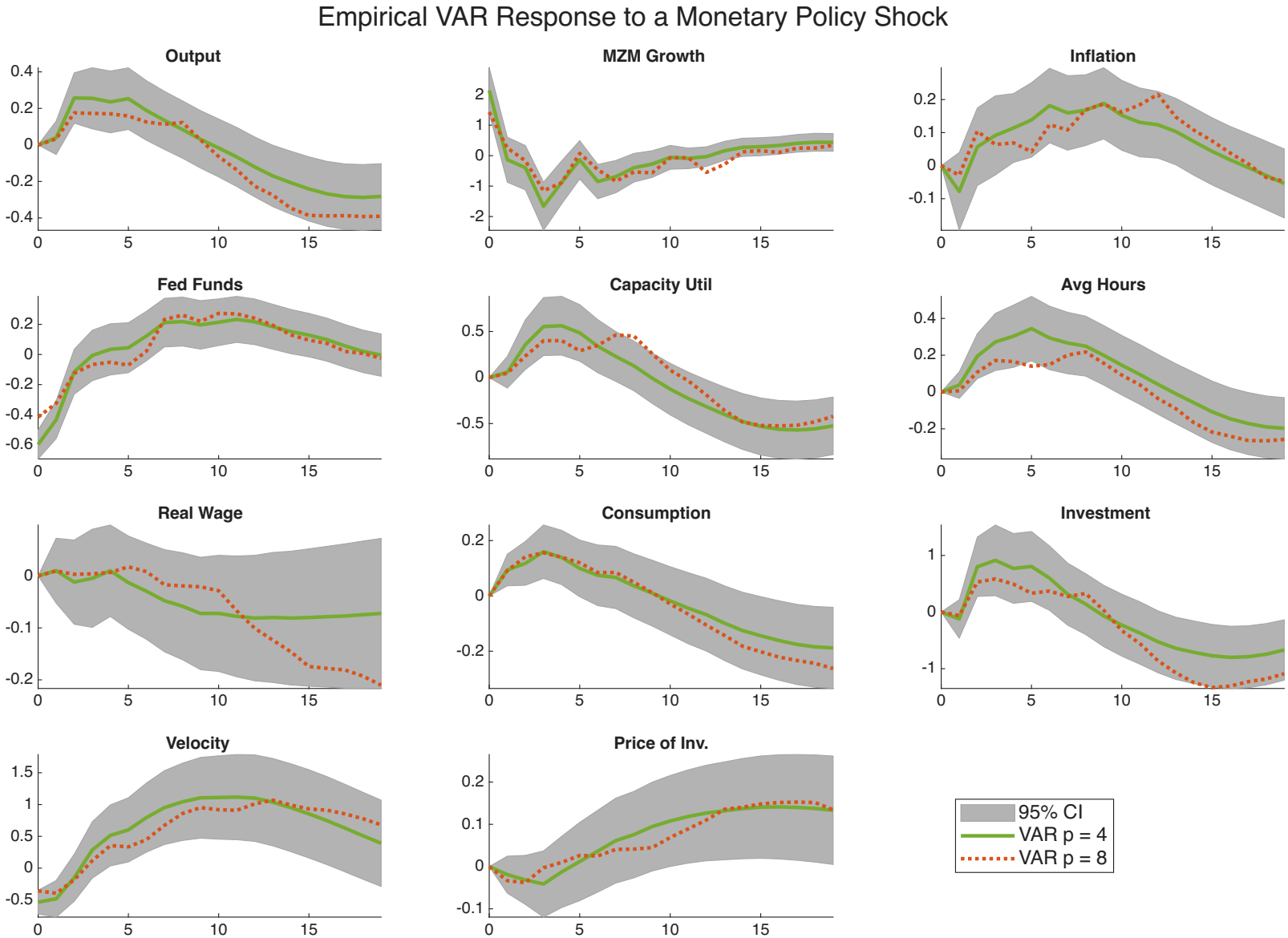


Figure 2: VAR IRFs Comparison: Monetary Policy Shock: 8 lags

VAR IRFs; Monetary Policy Shock

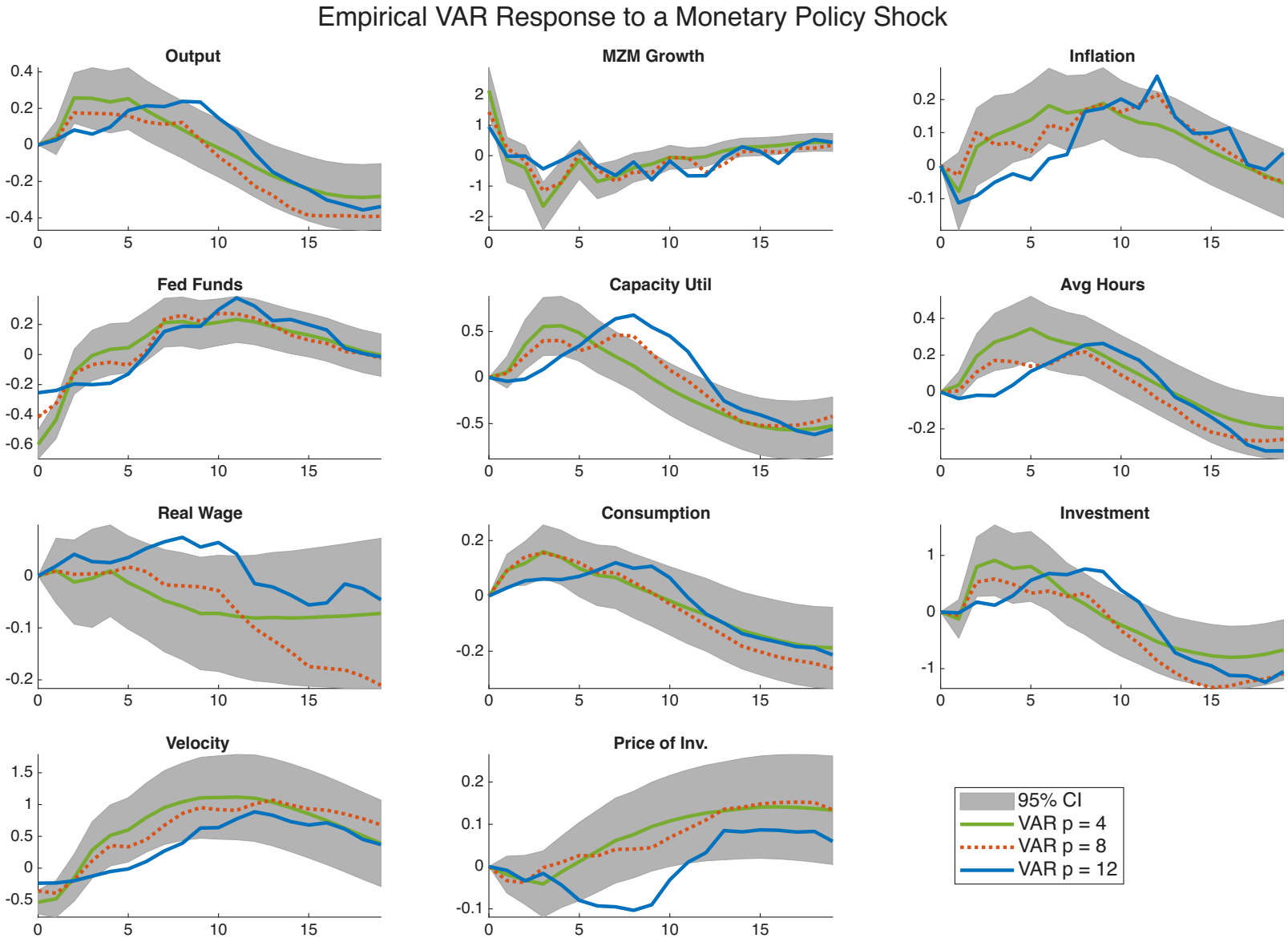


Figure 2: VAR IRFs Comparison: Monetary Policy Shock: 12 lags

VAR IRFs; Monetary Policy Shock

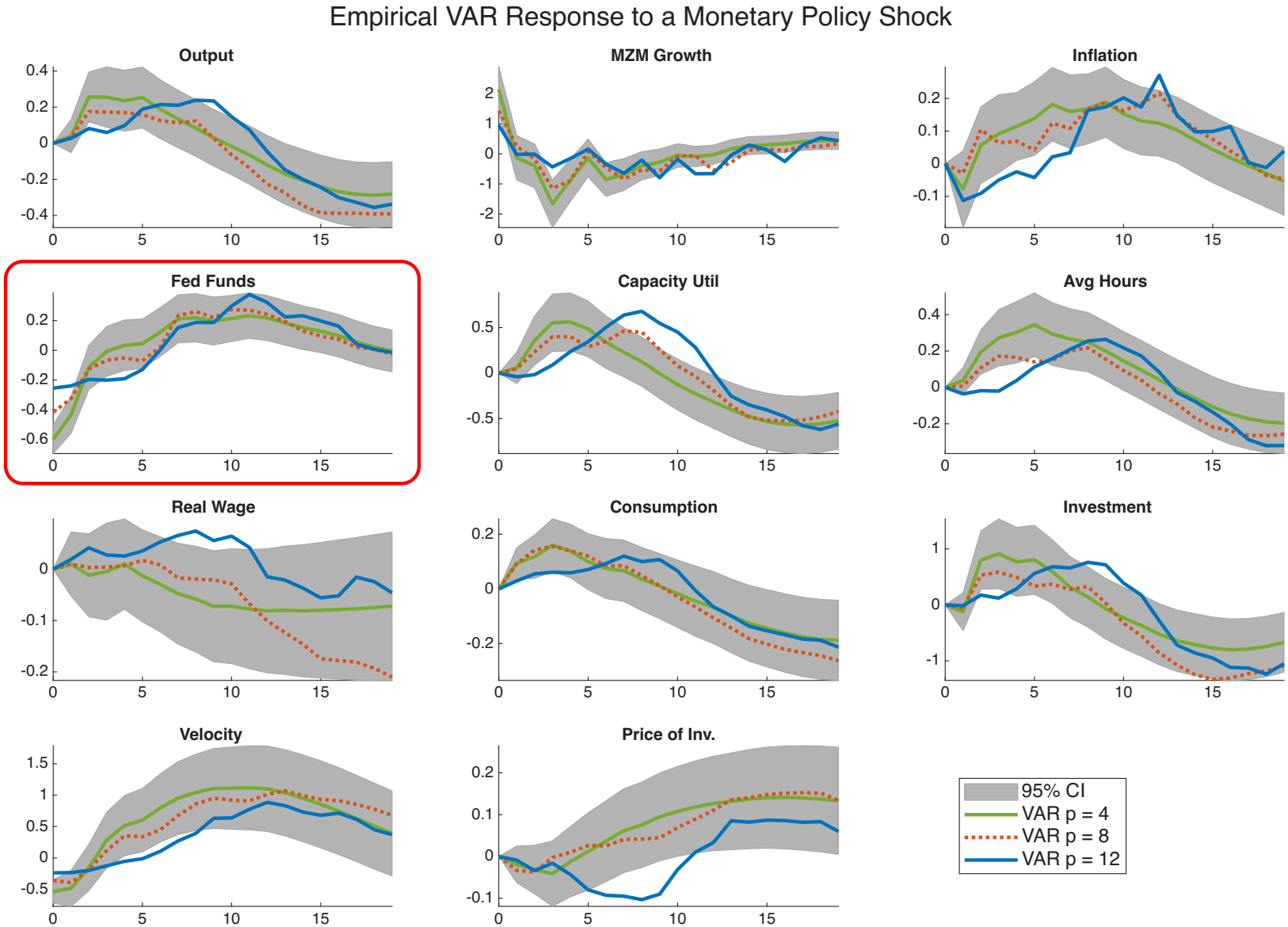


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VAR IRFs; Monetary Policy Shock

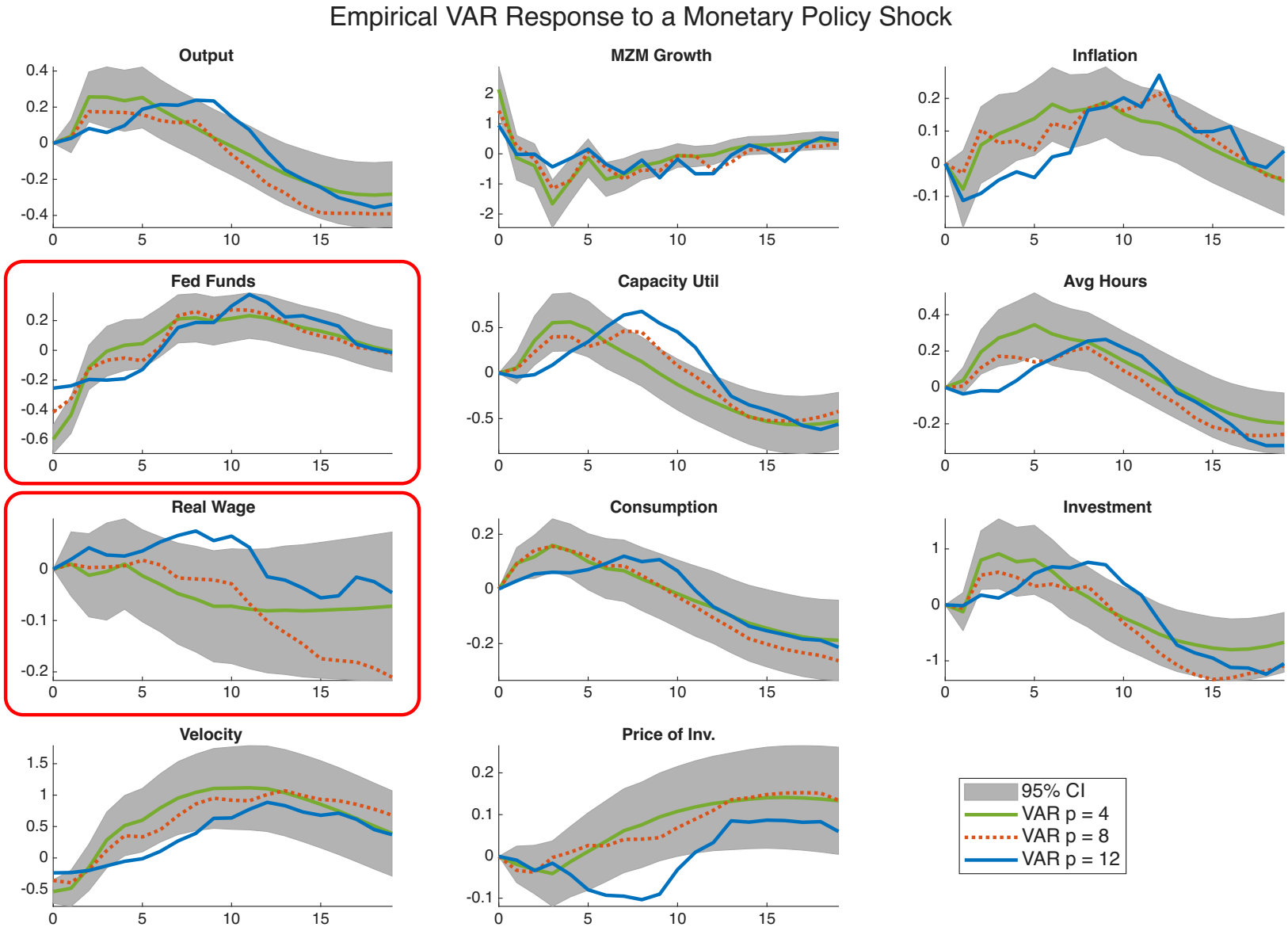


Figure 2: VAR IRFs Comparison: Monetary Policy Shock: 12 lags

VAR IRFs; Monetary Policy Shock

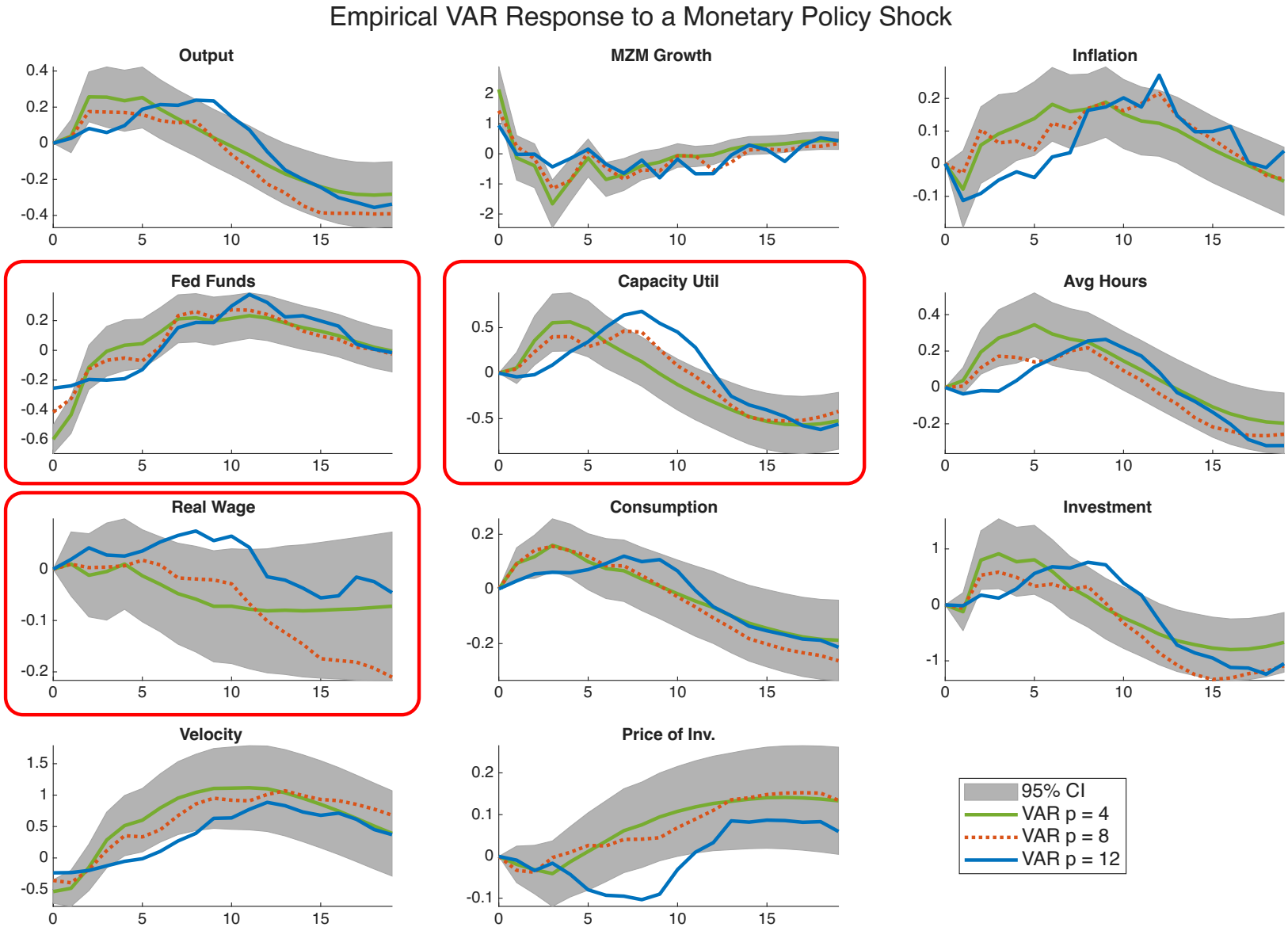


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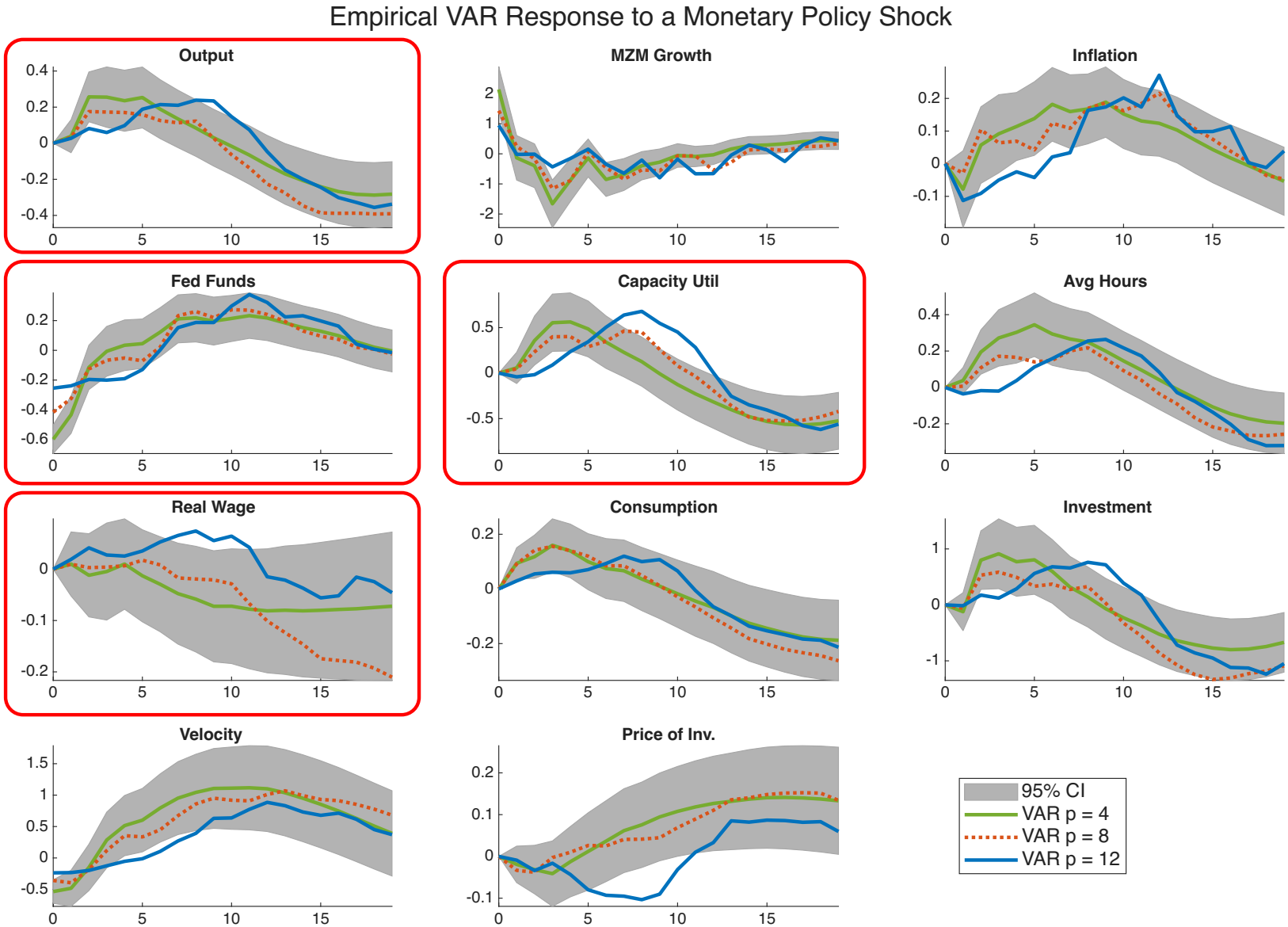


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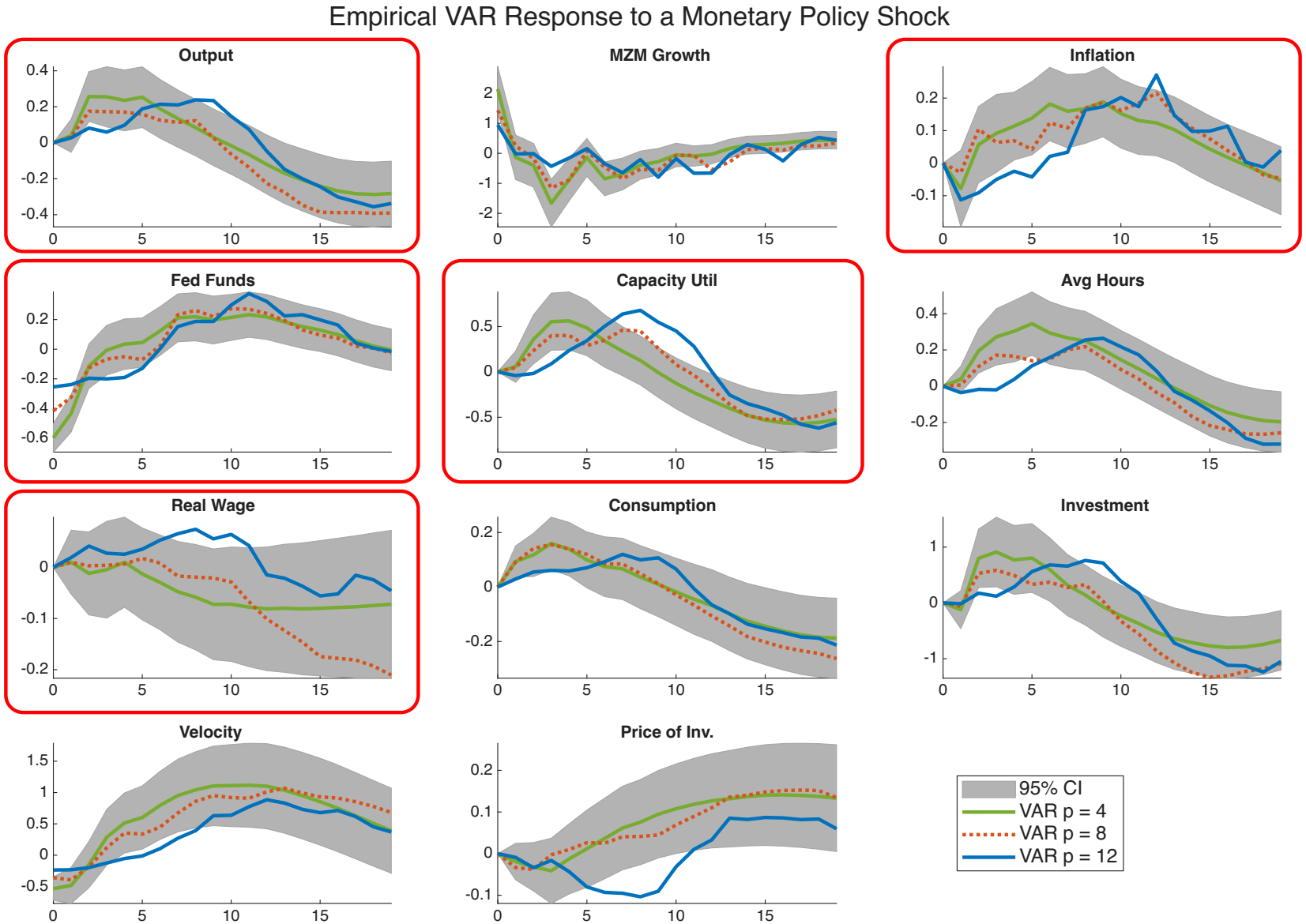


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DSGE Structural Parameters

MonteCalvo

Parameter	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Structural Estimated Frictions			
Price markup (λ_f)	1.0100	1.0100	1.0100
	(-)	(-)	(-)
Wage stickiness (ξ_w)	0.7221***	0.5617***	0.4838***
	(0.1506)	(0.1142)	(0.0757)
Phillips curve slope (γ)	0.0402	0.0402	0.0129***
	(0.0246)	(0.0598)	(0.0042)
Capacity util. cost (σ_a)	1.9938	0.1175	0.0042
	(2.6066)	(0.6299)	(0.0230)
Habit formation (b)	0.7066***	0.6935***	0.8596***
	(0.0463)	(0.0570)	(0.0203)
Inv. adj. cost (S'')	3.2807*	3.0372*	6.8431***
	(1.8874)	(1.5506)	(2.3306)
Money demand elast. (ϵ)	0.8077***	0.7531***	0.8346***
	(0.2111)	(0.1473)	(0.1944)
Implied Price Stickiness			
Homogeneous capital ($\xi_{p,hom}$)	0.8213***	0.8213***	0.8958***
	(0.0480)	(0.0574)	(0.0139)
Firm-specific capital ($\xi_{p,fs}$)	0.3372**	0.6448***	0.8884***
	(0.1505)	(0.1285)	(0.0220)

Note: Standard errors in parentheses. The Calvo parameters are recovered numerically from the estimated structural parameters; their standard errors come from a 1,000,000-draw Monte Carlo of the joint distribution of the estimated structural parameters.

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Implied Duration

Model Specification	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Homogeneous Capital Model			
Calvo parameter ($\xi_{p,hom}$)	0.82	0.82	0.90
Implied duration (quarters)	**5.6**	**5.6**	**9.6**
Firm-Specific Capital Model			
Calvo parameter ($\xi_{p,fs}$)	0.34	0.64	0.89
Implied duration (quarters)	**1.5**	**2.8**	**9.0**

Note: Implied duration is calculated as $D = 1/(1 - \xi_p)$.

- Firm-Specific model's Calvo Price parameter converges to the Homogeneous one
- Back to the micro-macro implied price duration mismatch

χ Across Lags

Conclusions

Following the advice of the Long-Lag VAR literature and replicating ACEL (2005) with an enriched VAR lag structure, I find that:

- Capacity Utilisation Cost, crucial variable to justify the firm-specific model, dramatically shrinks to 0
- Firm specificity of capital does not seem to be the key concept to explain inflation inertia
- ACEL's evidence on firm-specific capital is not robust to long-lag VARs

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De Graeve & Westermark (2025) and Baumeister (2025)

- Long-Lag VARs might have structural implications; it's up to the researcher to demonstrate that their findings are robust to an enriched lag structure

Thanks a lot for your attention!

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ACEL's MP Shock Propagation Mechanism

$\varepsilon_t^M \downarrow \Rightarrow i_t \downarrow \Rightarrow y_t^d \uparrow \Rightarrow y_t^s \uparrow \Rightarrow h_t^d \uparrow$
$\Rightarrow$ Calvo wages: $w_t \uparrow$, $\frac{w_t}{P_t} \uparrow$ little

Homogeneous

- $K_t = u_t \bar{K}_t \Rightarrow K_t^d \uparrow$ but $\bar{K}_t$ and $u_t \uparrow$ $r_t \uparrow$ given $\hat{\sigma}_a$
- $\Rightarrow MC_{i,t} = MC_t \Rightarrow$ MC independent of $y_{i,t}$ $\chi = 1$
- $MC_t \uparrow \Rightarrow P_t^{1-\xi_p} \uparrow \Rightarrow P_t \uparrow \Rightarrow \pi_t \uparrow$
- $\gamma = f(\bar{\xi}_p)$ given $\bar{\gamma}$ low, ξ_p needs to be high enough

Back to ACEL's DSGE Mechanism

ACEL's MP Shock Propagation Mechanism (Cont'd)

$\varepsilon_t^M \downarrow \Rightarrow i_t \downarrow \Rightarrow y_t^d \uparrow \Rightarrow y_t^s \uparrow \Rightarrow h_t^d \uparrow$
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Firm-Specific

- $K_{i,t} = u_{i,t}\bar{K}_{i,t} \uparrow \Rightarrow K_t^d \uparrow$ but $\bar{K}_t$
- Given $\chi < 1 \Rightarrow MC_{i,t} \uparrow$ in $y_{i,t}$
- $MC_{i,t} \uparrow \Rightarrow P_t^{1-\xi_p} \uparrow \Rightarrow P_t \uparrow \Rightarrow \pi_t \uparrow$
- $\gamma = f(\bar{\xi}_p; \bar{\sigma}_a)$ given $\bar{\gamma}$
- If $\sigma_a \to 0 \Rightarrow \chi \to 1 \Rightarrow \gamma = f(\bar{\xi}_p; 0) \Rightarrow \xi_{p,\text{FS}} \to \xi_{p,\text{H}}$

Back to ACEL's DSGE Mechanism

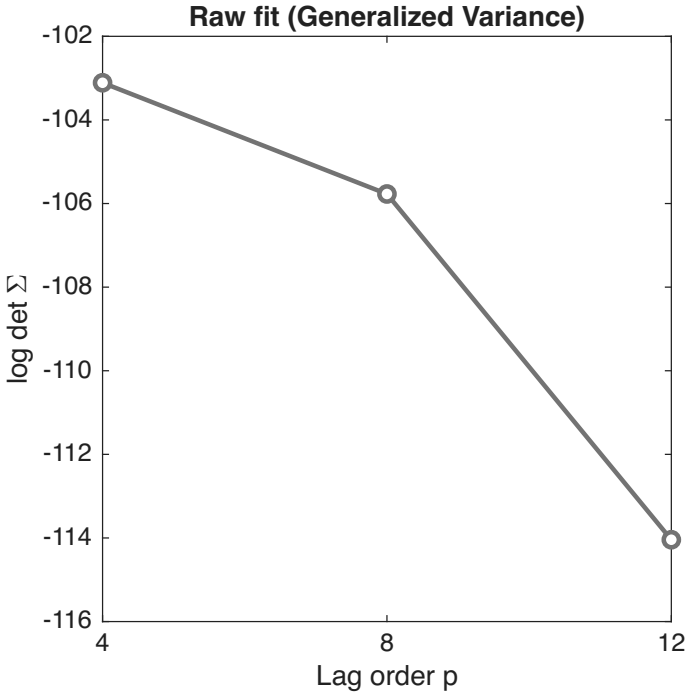
Implied Real Rigidity (χ) Across Lags

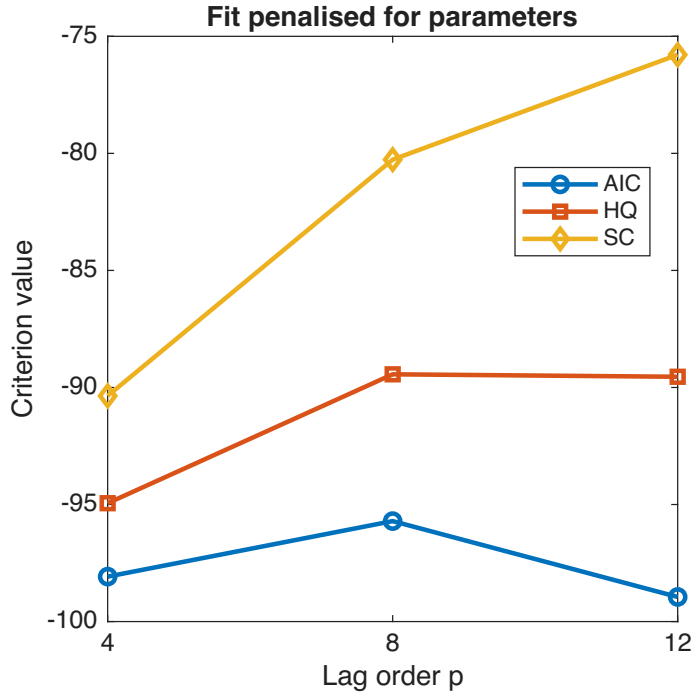
Model	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
χ_{hom} $(\equiv 1)$	1.0000 (-)	1.0000 (-)	1.0000 (-)
χ_{fs}	0.0307 (0.0596)	0.2030 (0.1667)	0.8671*** (0.1698)

Note: Standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. $\chi = \gamma\xi_p/[(1-\xi_p)(1-\beta\xi_p)]$, recovered by inverting the Phillips curve slope equation; $\chi_{hom} \equiv 1$ by construction. Standard errors for χ_{fs} come from the same 1,000,000-draw joint Monte Carlo used for the Calvo parameters.

Back to Implied Duration

Generalised Variance and Information Criteria





Lag	$\log\det\hat{\Sigma}$	**AIC**	**HQ**	**SC**
$p = 4$	-103.1111	-98.0796	**-94.9444**	**-90.3591**
$p = 8$	-105.7716	-95.7087	-89.4383	-80.2677
$p = 12$	-114.0426	**-98.9483**	-89.5426	-75.7867

Note: Bold marks the lag order each criterion selects (lowest value).
Selected lag: AIC $\to p = 12$, HQ $\to p = 4$, SC $\to p = 4$.

Back to Residuals' Variance Reduction

VAR IRFs All CI; Monetary Policy Shock

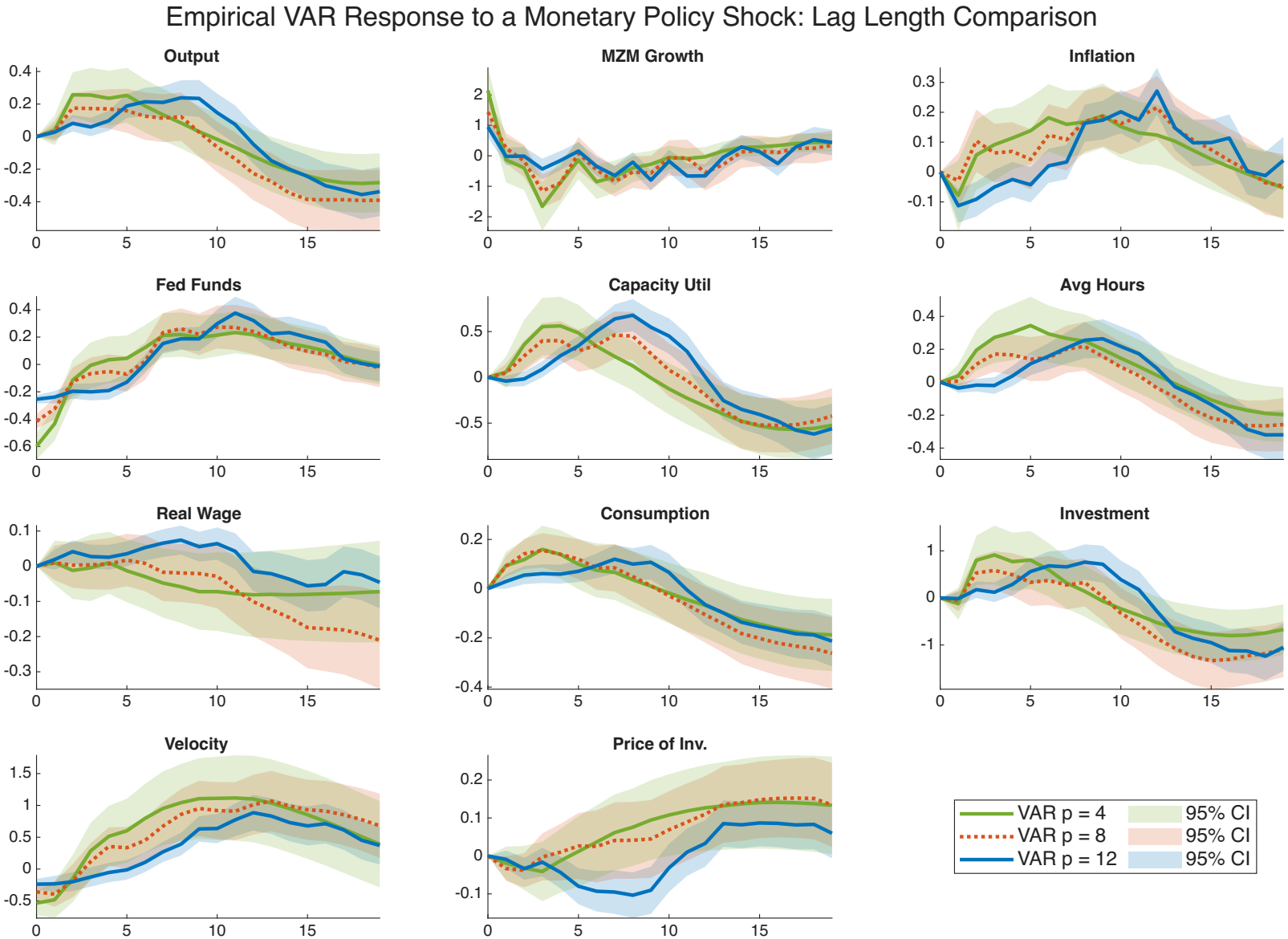


Figure 3: VAR IRFs Comparison with all CIs: Monetary Policy Shock

Bootstrap Confidence Intervals Shrink

- Compared to the examples in De Graeve and Westermark (2025), this exercise uses a medium-scale VAR: 10 variables + constant
- Estimating the VAR at $p = 12$, we approach the feasible limit: only $T_u - k = 38$ degrees of freedom per equation, and a total number of coefficients $n + n^2 p = 1210$
- Bootstrap Confidence Intervals actually shrink
- Confirming De Graeve and Westermark (2025) p.2 JME point on variance reduction even in a limit case "Misspecification reduction not only leads to Bias-Variance trade-off, but it reduces variance"

Identification Bias

Ravenna (2007) finds a second channel through which lag truncation operates: **Identification Bias**

- It is a **different source of bias** from pure truncation
- It arises even when the identifying assumptions are correct, and it emerges in population
- **It appears when the AR coefficients enter the identifying map's computations of structural shocks**
- If the AR coefficients are biased, the identifying map is biased as well

- It is not present in short-run Cholesky identification
- It is present in long-run restrictions identification à la Blanchard and Quah (1989)

Back to Truncation Bias

Long-Run Restrictions' Fragilities

Using the Blanchard and Quah (1989) identification as an example.
It is a long-run Cholesky that restricts the cumulative effect of a shock on the level of a variable to be zero, which is equivalent to restricting the sum of the impulse responses to be zero.

A VAR(p) does not estimate the sum consistently when the true representation is of infinite order, therefore the restrictions are imposed on the wrong object.

This has consequences for where the bias might appear. A long-run restriction says nothing about any finite horizon. Therefore the bias can appear at any horizon, impact included, not just the long run.

Back to Truncation Bias

VAR IRFs; Neutral Technology Shock

Back to MP Shock

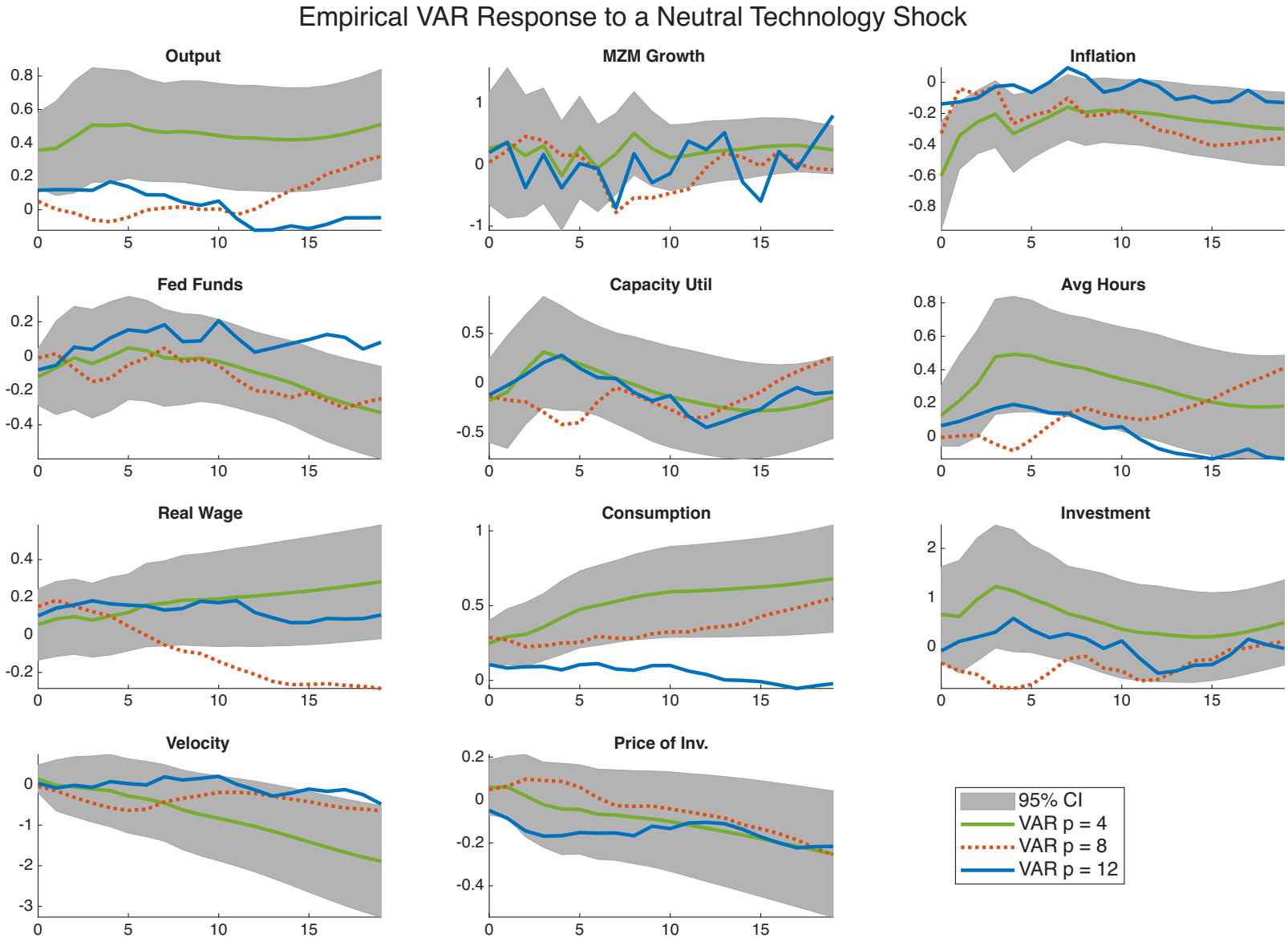


Figure 4: VAR IRFs Comparison: Neutral Technology Shock

VAR IRFs; Embodied Technology Shock

Back to MP Shock

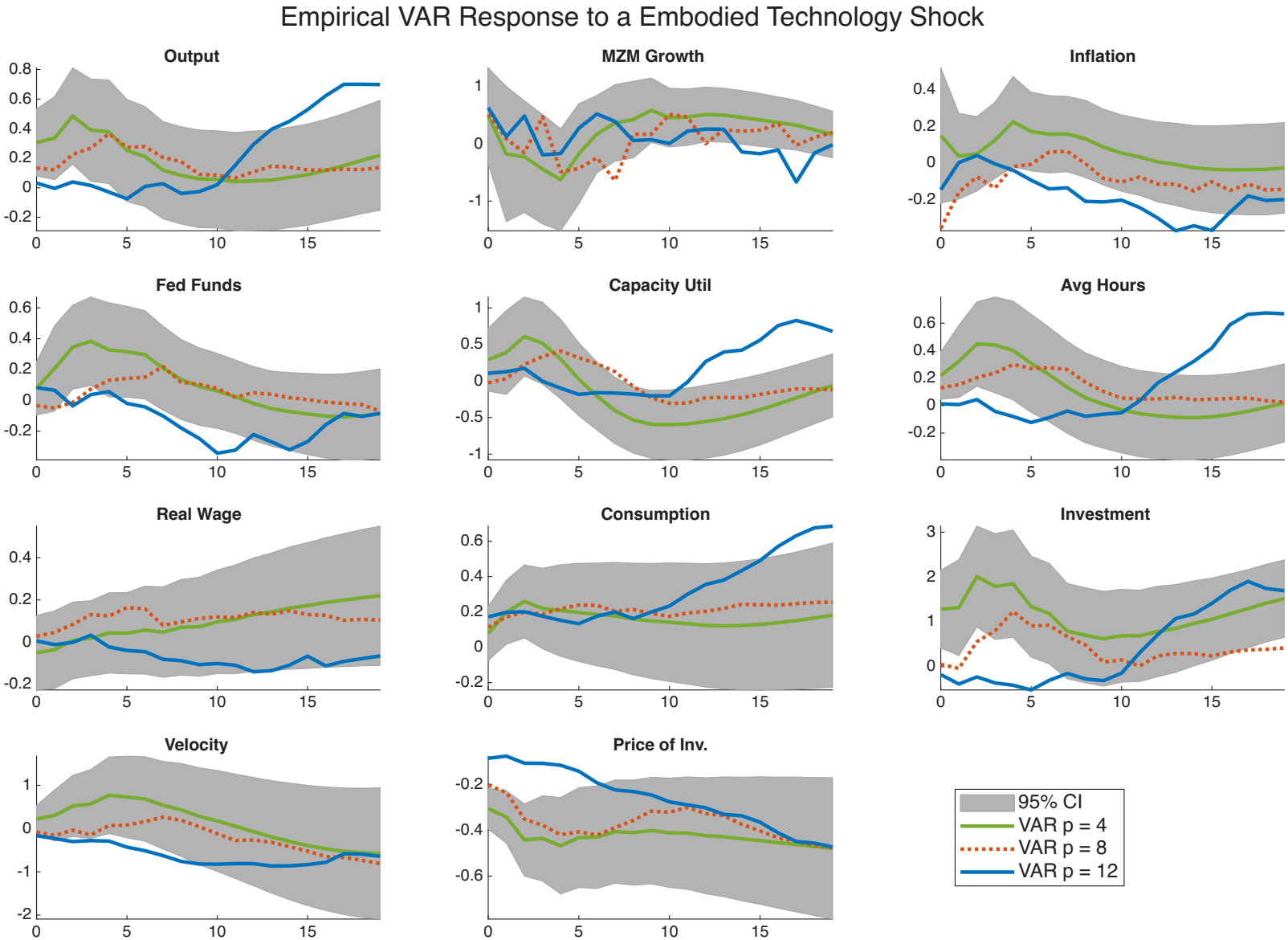


Figure 5: VAR IRFs Comparison: Embodied Technology Shock

Tech Shocks Comment

Back to MP Shock

- The Identification Bias is binding, and its effect can be seen in the fact that the responses disagree across lags at every horizon
- Under short-run recursive identification, most impact responses to an MP shock are zero by construction. Pure truncation bias distorts the propagation of a correctly isolated shock, so it accumulates with the horizon and cannot by itself generate a disagreement at $h = 0$
- MP shock responses have kept their sign and shape, moving only in timing
- Tech shock responses, identified by long-run restrictions, do not agree at any horizon, including the impact, and the disagreement is not only in timing but also in sign

DSGE IRFs; Monetary Policy Shock

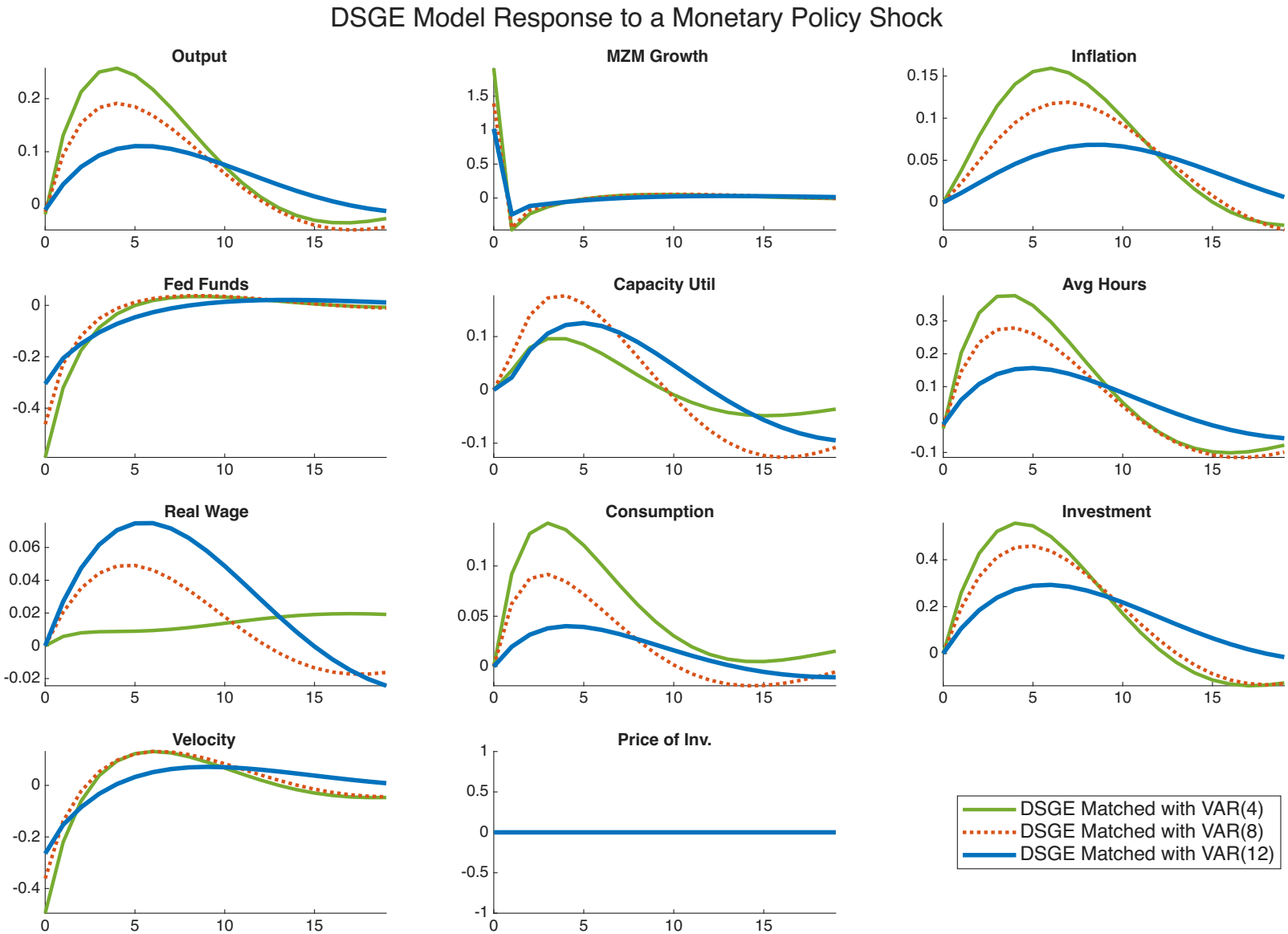


Figure 6: DSGE IRFs Matched with VAR(4), VAR(8) and VAR(12): Monetary Policy Shock

MonteCalvo

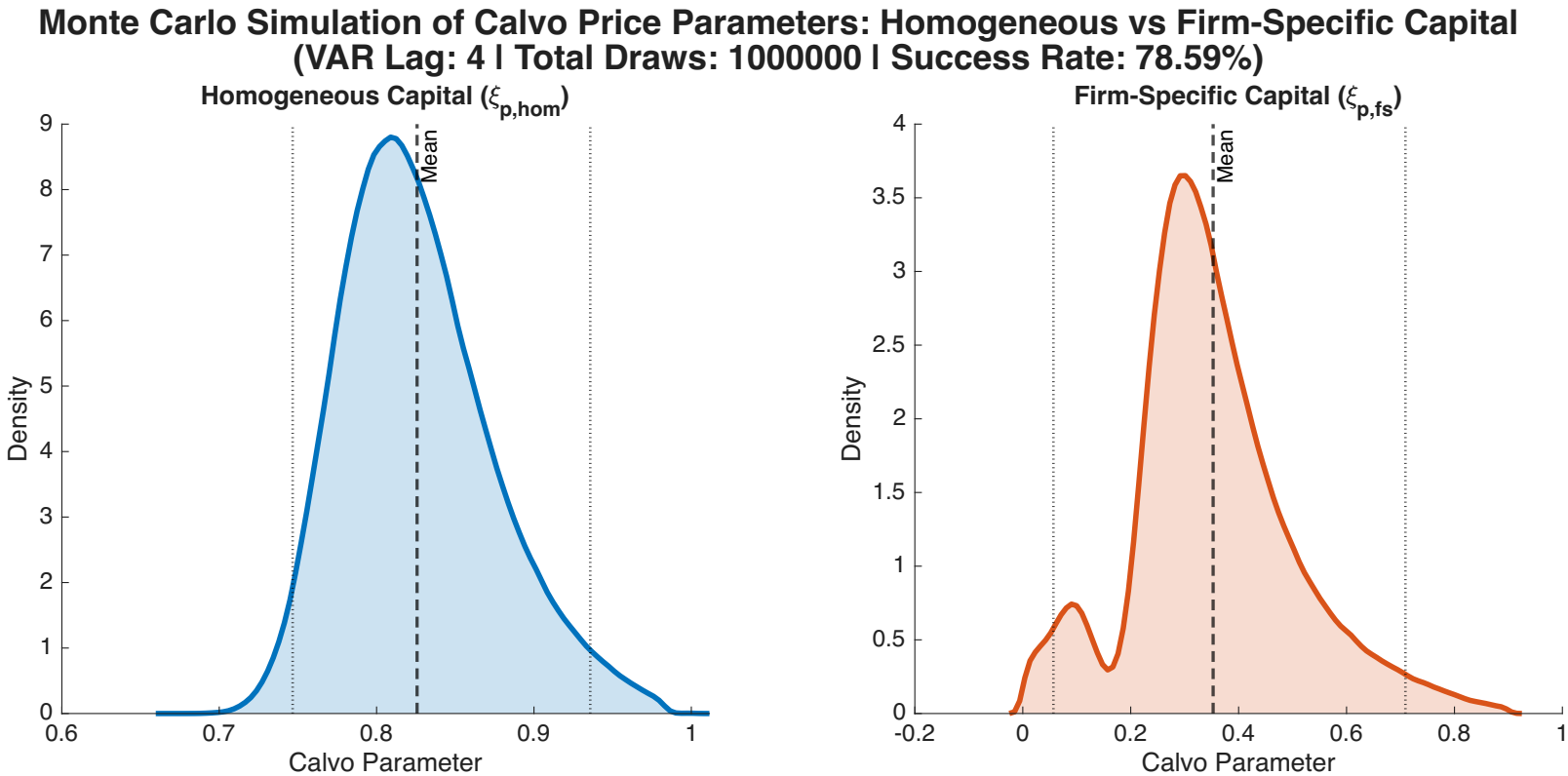


Figure 7: Calvo Price Parameter PDF from Monte Carlo Simulation; VAR(4)

Back to DSGE Parameters

MonteCalvo

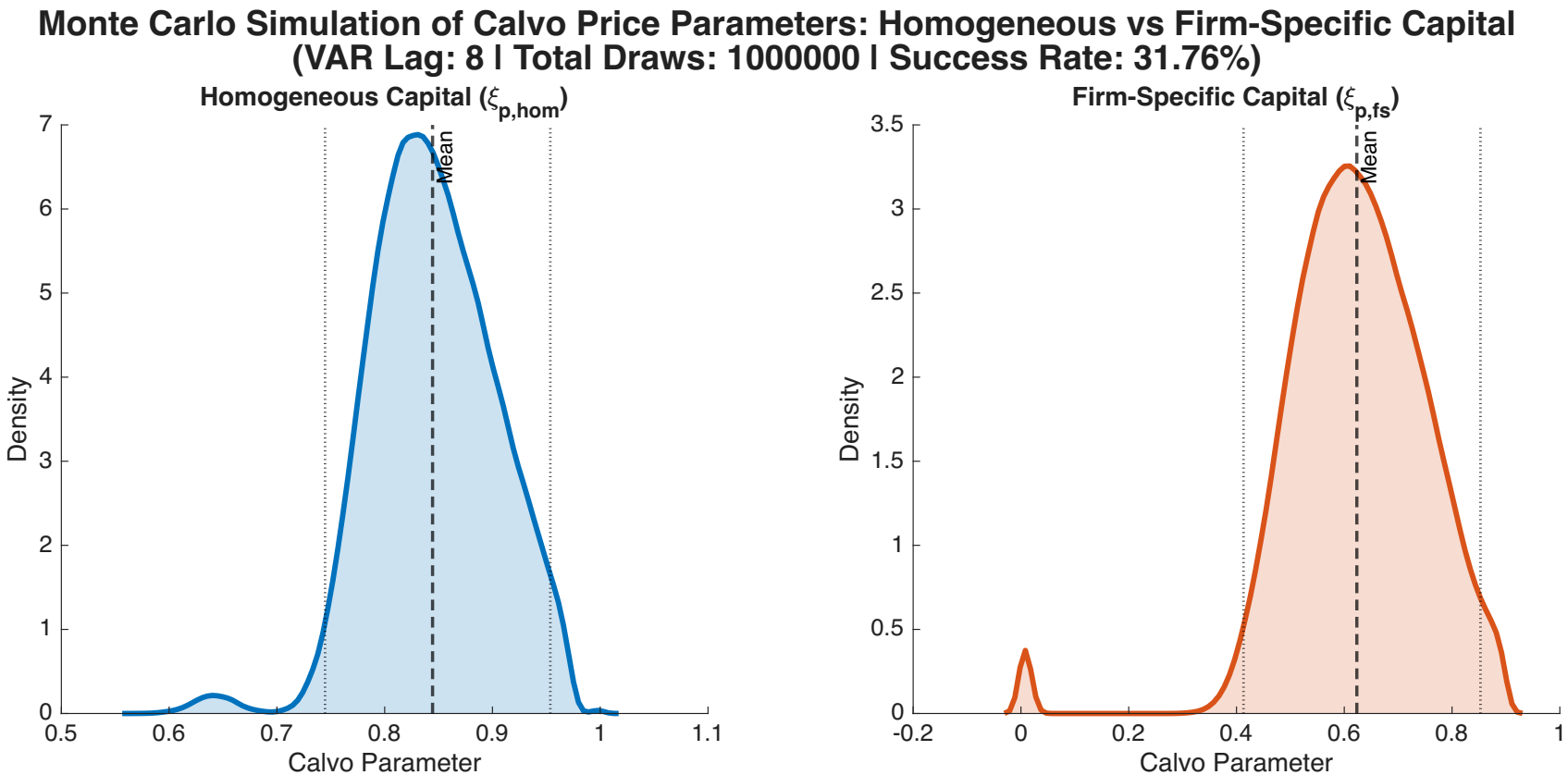


Figure 8: Calvo Price Parameter PDF from Monte Carlo Simulation; VAR(8)

Back to DSGE Parameters

MonteCalvo

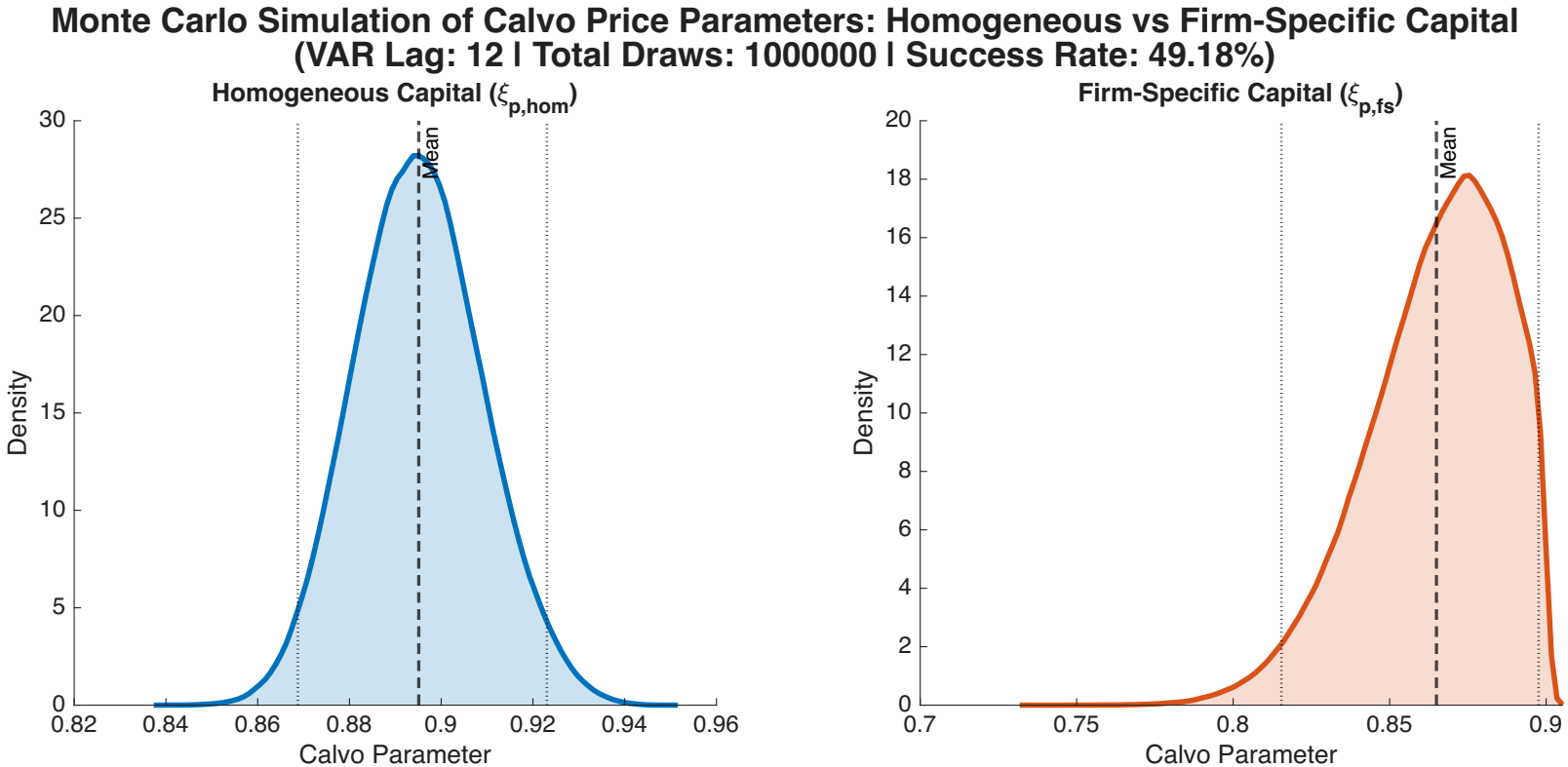


Figure 9: Calvo Price Parameter PDF from Monte Carlo Simulation; VAR(12)

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